

Smart Outcome Predictor - Final Report

Objective

The objective of this project is to build an ensemble based machine learning system to predict student course completion and final performance score.

Dataset Description

The dataset contains 5200 student records with demographic information, engagement metrics, assessment scores and two target variables: completion_status and final_score.

Data Cleaning Summary

The student_id column was removed. Missing values in time_spent_hours, avg_quiz_score and attendance_rate were filled using median imputation. Duplicate rows were checked and removed.

EDA Summary

Exploratory analysis showed that students with higher time spent, attendance rate and quiz scores are more likely to complete the course and score higher. The dataset is slightly imbalanced towards non completion.

Models Used

Bagging, AdaBoost, Gradient Boosting, LightGBM, XGBoost, Voting Classifier (hard and soft) and Stacking were implemented for classification and regression tasks.

Evaluation Results

Classification models were evaluated using Accuracy, Precision, Recall, F1 Score and ROC-AUC. Regression models were evaluated using MAE, MSE, RMSE and R2 Score. Detailed results are saved as CSV files in the report folder.

Best Classification Model: Gradient Boosting

Best Regression Model: Gradient Boosting

Bagging vs Boosting

Bagging trains models in parallel and mainly reduces variance, making predictions more stable. Boosting trains models sequentially, focusing on previous errors, and mainly reduces bias, leading to higher accuracy but slightly more risk of overfitting.

Comparison of Tree-based Boosting Algorithms

AdaBoost is simple but sensitive to noise. Gradient Boosting gives good accuracy but trains slower. LightGBM trains faster using leaf-wise growth. XGBoost generally gives the most robust results due to its built-in regularization.

Advantages of Voting

Voting combines different types of models, reducing the impact of a single model's mistake and improving overall stability of predictions.

Advantages of Stacking

Stacking uses a meta-learner to learn the best way to combine base models, often resulting in better performance than any individual base model.

Business Interpretation

The platform can use the classification model to identify students at risk of not completing the course and the regression model to predict expected performance, allowing timely intervention and support.

Final Recommendation

It is recommended to deploy the best performing ensemble models identified in this project, since they provide more stable and accurate predictions compared to a single base model.

Final Conclusion

This project successfully demonstrates that ensemble learning methods such as Bagging, Boosting, Voting and Stacking improve prediction accuracy and stability for both classification and regression tasks in an EdTech setting.